F12 4th CAR

Lecture 13, Section 2.1
Quadratic Functions and Models

Quadratic function

Parabola

Standard form of a quadratic function

Axis of symmetry

Graphing a quadratic function

Practice

① Find the inverse of $f(x) = \sqrt{2x-3}$

② Graph f and f^{-1}.

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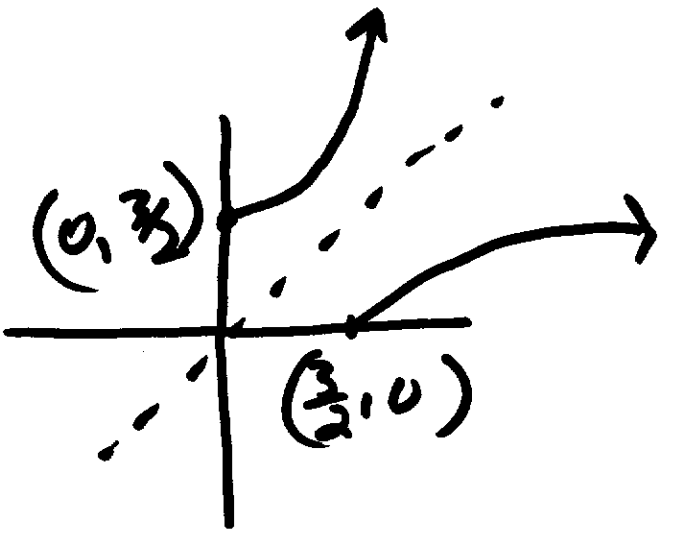
sol. ① $f^{-1}(x) = \dfrac{x^2+3}{2}, \quad x \geq 0$



Lecture 13, Section 2.1
Quadratic Functions and Models

<u>**Def.**</u> A **quadratic function** is a second degree polynomial of the form

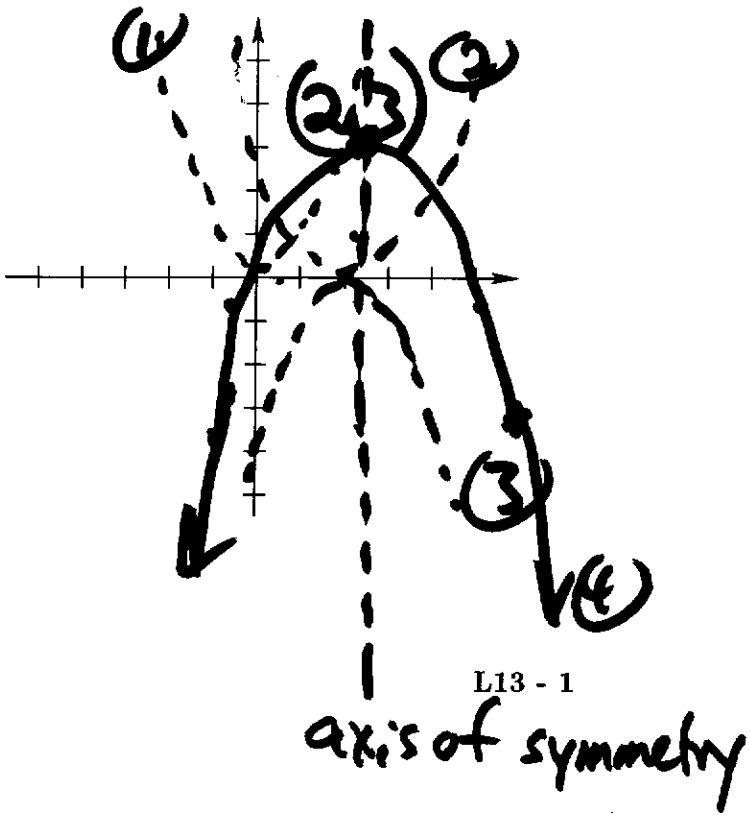
$$f(x) = ax^2 + bx + c$$

general form

where a, b and c are real numbers and $a \neq 0$.

NOTE: The graph of a quadratic function is a transformation of the parent function $f(x) = x^2$. Its graph is in the shape of U called a **parabola**.

<u>**ex.**</u> Graph $f(x) = -(x-2)^2 + 3$



(1) $y = x^2$

(2) $y = (x-2)^2 \rightarrow 2$

(3) $y = -(x-2)^2$ refl. across x-axis

(4) $y = -(x-2)^2 + 3$

$\uparrow 3$

Standard form of a quadratic function

The quadratic function can be written in **standard form**

$$* \quad f(x) = a(x - \underline{h})^2 + \underline{k}, \quad a \neq 0$$

The standard form is convenient for sketching:

1. **Vertex:** (h, k)

2. **Axis of Symmetry:** line $x = h$

All parabolas are symmetric with respect to a line called **axis of symmetry**.

3. Parabola opens upward if $a > 0$; the vertex is the minimum point on the graph.

Parabola opens downward if $a < 0$; the vertex is the maximum point on the graph.

4. The graph is a transformation of $f(x) = x^2$.
If $|a| > 1$, the graph is **stretched vertically** and if $0 < |a| < 1$, the graph is **compressed vertically**.

$f(x) = a(x-h)^2 + k$

$\left(\frac{1}{2}\cdot 4\right)^2 = 2^2$
$= 4$

To write a quadratic function in standard form, we **complete the square**.

<u>ex.</u> Write the function $f(x) = 2x^2 + 8x + 7$ in standard form and graph.

① Write in general form

② Factor out a from the x terms.

③ Complete the square.

④ Move the outside of the parentheses

⑤ Finish.

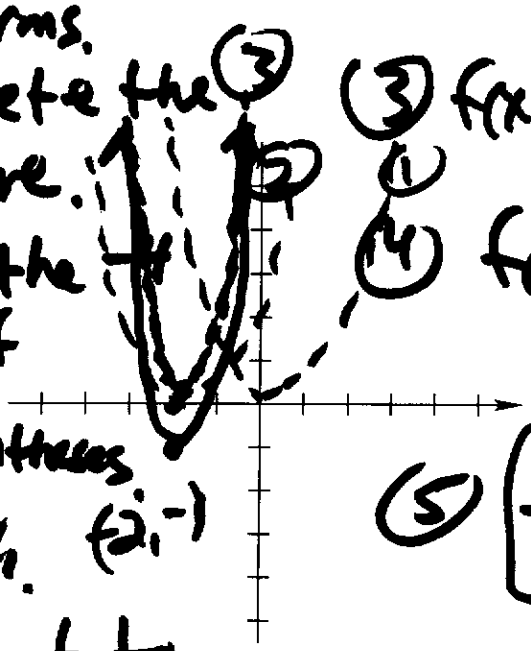
① $f(x) = 2x^2 + 8x + 7$

② $f(x) = 2(x^2 + 4x\ \ ) + 7$

③ $f(x) = 2(x^2 + 4x + 4 - 4) + 7$

④ $f(x) = 2(x^2 + 4x + 4) + 7 - 8$

⑤ $\boxed{f(x) = 2(x+2)^2 - 1}$



* Short-cut to finding the vertex.

① $y = x^2$

② $y = (x+2)^2 \leftarrow -2$

③ $y = 2(x+2)^2$ stretch vert.

④ $\downarrow 1$

NOTE: If we start with equation $f(x) = ax^2 + bx + c$, we can complete the square to write in standard form $f(x) = a(x-h)^2 + k$ with

$$h = -\frac{b}{2a} \text{ and } k = f(h) = \frac{4ac - b^2}{4a}.$$

x-coor of vertex

y-coor of vertex

not required to memorize

$y = -x^2 + 6x - 8$

Find intercepts to help sketching

To find x-**intercept** of the graph of f, we set $y = 0$ and solve the equation $f(x) = 0$.

To find y-**intercept**, we set $x = 0$ and we have $y = f(0) = c$.

ex. For each quadratic function, find the vertex, axis of symmetry, intercepts, domain and range. Graph the function.

1) $f(x) = -x^2 + 6x - 8$

(1) $-\frac{b}{2a} \rightarrow \frac{-6}{2(-1)} = 3$

$f(3) = -9 + 18 - 8 = 1$

$\boxed{V(3,1)}$

(2) axis of sym $\boxed{x=3}$

(3) x-int, let $y=0$

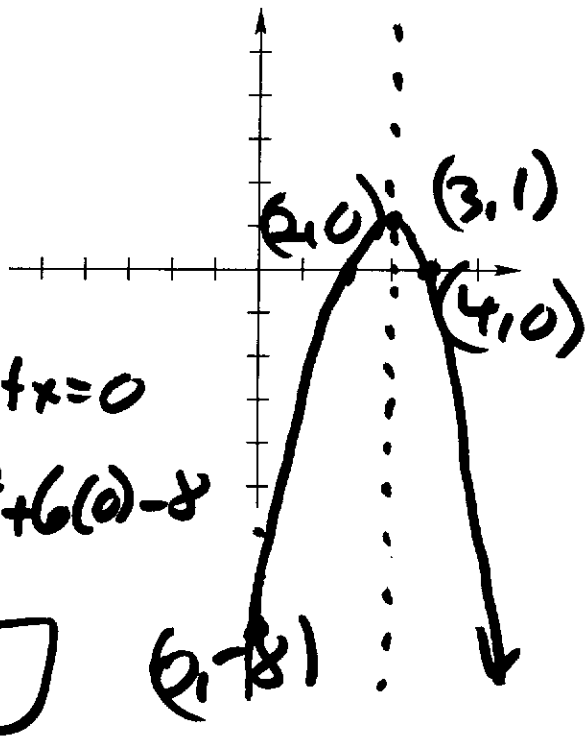
$-x^2 + 6x - 8 = 0$

$x^2 - 6x + 8 = 0$

$(x-4)(x-2) = 0$

$x=4 \quad x=2$

$\boxed{(4,0) \quad (2,0)}$



(3) y-int, let $x=0$

$y = -(0)^2 + 6(0) - 8$

$y = -8$

$\boxed{(0,-8)}$

(4) $\boxed{D: (-\infty, \infty) \quad R: (-\infty, 1]}$

① vertex $\frac{-b}{2a} \rightarrow \frac{-2}{2(4)} = -\frac{1}{4}$ $\quad f(-\frac{1}{4}) = 4(-\frac{1}{4})^2 + 2(-\frac{1}{4}) + 1$

② Axis of sym. $\boxed{x = -\frac{1}{4}}$ $\quad = \frac{3}{4}$

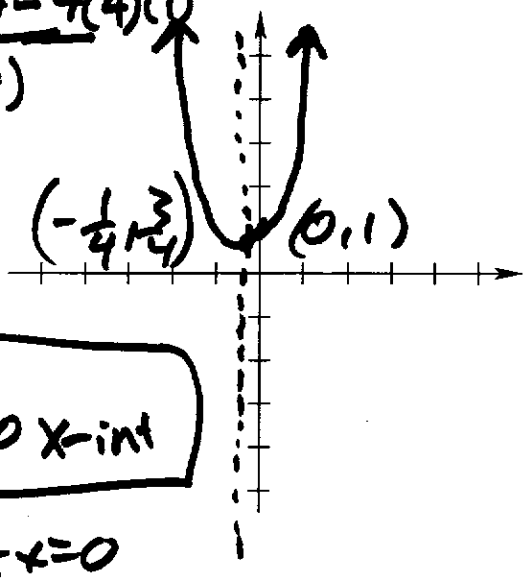
③ x-int, let y=0 $\quad$ thus $\boxed{V(-\frac{1}{4}, \frac{3}{4})}$

$4x^2 + 2x + 1 = 0$

2) $g(x) = 4x^2 + 2x + 1$

$x = \frac{-2 \pm \sqrt{4 - 4(4)(1)}}{2(4)}$

$x = \frac{-2 \pm \sqrt{-12}}{8}$



$\boxed{\text{no real sol., so no x-int}}$

$\boxed{\begin{array}{l} D: (-\infty, \infty) \\ R: [\frac{3}{4}, \infty) \end{array}}$

y-int, let x=0

$g(0) = 1$

$\boxed{(0,1)}$

Write the equation of a parabola

<u>ex.</u> Find the equation of the parabola with vertex $(1, 2)$ which passes through the origin.

we know $(1,2)$, also $(0,0)$ is on the graph.

$f(x) = a(x-h)^2 + k$ sub in

$y = a(x-h)^2 + k$

$0 = a(0-1)^2 + 2$

$0 = a + 2$

$a = -2$

thus $\boxed{f(x) = -2(x-1)^2 + 2}$

Applications

<u>**ex.**</u> Suppose that a ball is thrown straight up by a student sitting on a wall that is 6 feet high with initial velocity of 64 feet per second. The function

$$h(t) = -16t^2 + 64t + 6$$

expresses the height of the ball in feet as a function of the time t in seconds, with $t = 0$ corresponding to the instant when the ball is released.

1) At what time will the ball reach its maximum height? Graph is upside down parabola, thus it will have a maximum value (height).

Find the vertex. $\frac{-b}{2a} \Rightarrow \frac{-64}{2(-16)} = \boxed{2 \text{ seconds}}$

2) What is the maximum height of the ball?

$h(2) = -16(2) + 64(2) + 6 = \boxed{70 \text{ feet}}$

Note: $p =$ price
$x =$ # sold

<u>ex.</u> The unit price p of a certain product and quantity sold, x are related by the **demand equation**

price $= p = -\frac{1}{2}x + 30, \ 0 \le x \le 60.$

1) Express revenue R as a function of x. money you put in the cash register.

Revenue = (price)(# sold) = $p \cdot x$

$R(x) = \left(-\frac{1}{2}x + 30\right)x = \boxed{-\frac{1}{2}x^2 + 30x = R}$

2) What quantity x will maximize revenue? What will the maximum revenue be?

$-\frac{b}{2a} \rightarrow \frac{-30}{2(-\frac{1}{2})} = \boxed{\text{30 units will max. the revenue}}$

$R(30) = -\frac{1}{2}(30^2) + 30(30) = \boxed{\$450}$

3) What **price** should be charged to maximize revenue? Found out we should sell 30 units

$p = -\frac{1}{2}x + 30$

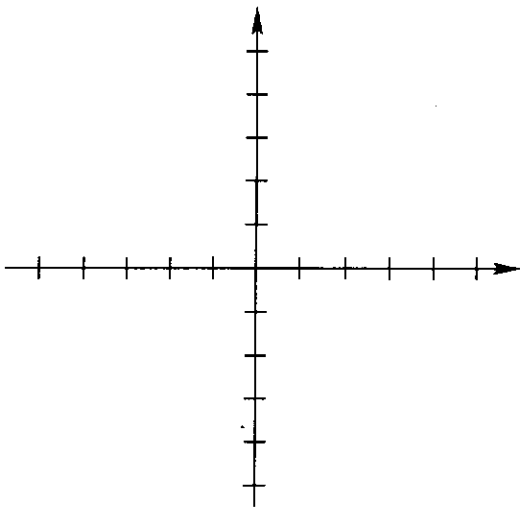
$p(30) = -\frac{1}{2}(30) + 30 = \boxed{\$15}$

or $R = p \cdot x$

$450 = p(30) \longrightarrow \boxed{p = \$15}$

Practice.

1) Write the function $f(x) = 3 - 4x - 4x^2$ in standard form and graph.



2) Find the quadratic function that has $x-$intercepts 1 and 5 and $y-$intercept 10.

Answer. 1) $f(x) = -4\left(x + \frac{1}{2}\right)^2 + 4$ 2) $y = 2x^2 - 12x + 10$

Pop-lecture Quiz 3

Name _____

TA _____

Sect _____

Given the one-to-one function $f(x) = \dfrac{x+1}{x-2}$

a) find its inverse.

b) find the range of f.